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Hello? Hi Berit, how are you? I'm good, and you? How are you? Sorry, we left the meeting super fast because we finished really fast. I saw that you arrived and no one was there. Yeah, no, it's okay.

I was waiting for Jean-Philippe. He told me that he would send the link this morning. Then I didn't hear back, so I wanted to come to the school and I was on my way.

I'm sorry for this. It takes a long time. When you need it, that's the time that it picks to restart and update and all of that.

Yes, exactly. The one that you really most like is the one that, oh, I'm going to restart. Exactly.

I totally get it, no worries. How are you? How is the project going? Great, great. Actually, I'm, you know, I have my exact same environment up and running.

I can control the robot, send the sensor pad where I want, collect the data, which is all good. But I have some problems that I thought maybe I can get your idea on that. Sure.

Yeah, first of all, when I have my, your example is a cylinder, right? The one that, yeah. So, when I have the, let's say this is the sensor pad and this is the cylinder. So, when I have a cylinder like this, I get a beautiful vertical contact.

But when I have it tilted, I'm sorry, this one is too big, but maybe, yeah. When I have the cylinder tilted like this for 20 or 35 or 45 degree, I don't get the tilt. I don't get the angle in the tactile data.

And then I went back and I read your paper again and I realized that you had the sensor fixed and you just pressed different indenters to the sensor. Yes. So, could we say that your training, your CNN is not being, was not trained on angles or tilted contacts? No, yes, it was training for that.

The problem is like the way maybe when you put the object, sometimes it's like, that's the problem with Isaac. Sometimes it's like moving the body or like the, I don't know how to say, like, there is a problem, Isaac, when you put the object like completely straight, there is not so much problem. But when you keep the object like in a certain position, you need to make it kinematic.

I don't know right now if the object is kinematic. I did made it kinematic. Yeah, I did that.

It doesn't move and I can see the way it renders the environment. It doesn't move. And I tried so many different angles.

Something else that I tried is that I realized that you made the object with like a scale. So, if I change the scale, I can get different diameter for a cylinder. So, I changed the diameter of the cylinder, like made it bigger and smaller.

Still nothing changed. And another thing was that the closure angle for like the current example is 0.55. So, I made it even bigger. So, I make sure that the sensor are pressing hard against the object.

Yes. But when I have, as I said, when I have a vertical cylinder, it's beautiful and clean. Even when it is like a vertical, I have higher pressure on the top.

And as it goes down, is less pressure on the bottom. Yeah. Could it be because of the hinges? As we are closer to the hinges, we get more pressure and the tip of the finger is not pressing that hard.

Could it be that? Yeah, it's something related with like, exactly with the gripper tool. They mentioned the sensor. Honestly, I didn't see that.

But I'm going to check for the angle. If you have like your script, could you share with me? And I could check. Sure.

If you have seen the script, that could be better. And I could check. Otherwise, I'm really like trippy, like share with you.

Like now we are moving completely to use like a real like basic engine. And it looks more stable. We can control the point that we are applying the force.

I'm sorry, what are you moving toward? What was that? We are moving to a different like, we are going to use Isaac. But now we are not going to complete the pencil. Isaac, we are going to use Newton.

Uh-huh. Like it's kind of like working like basically based on Muchoco. So, there the sensor is working really nice.

Like even we can detect the force that you are applying to the body. Or the force that you are applying to the sensor. At one point, we are going to be able to control the gripper based on the force that you are applying.

Like to the object like it's happening with a real gripper. Things that is not happening right now with Isaac. But that maybe is going to be working like at the end of August.

Okay. Yeah, it would be nice. But I'm pretty tight on my schedule.

So, I'm going to go ahead and work with this. But yeah, I definitely would be excited to use the Newton version as well. But for now, I guess I have to deal with this and try to get some data.

Yes. And you know, this way I can complete the pipeline. But down the line, maybe end of August or whenever you have the other version.

If I can use that, I would be happy to replace this. Yeah, for sure. And get more data.

Yeah, really. For you to say to me like I don't have any idea because the network is controlled. I tested like with different positions.

The one I put in there is just like the most beautiful one. But you can share me your files. I can check.

Yeah, I'm going to share my script and some like snapshots and results. So, you know how the design of the scene was and the tactile data that I got. So, you tell me if that makes sense or not.

Perfect. Anyway, I need to do like a final update to the record. And it could be nice if I see the mistake before and I can like update there like not to have the same problem all the time.

Sure. Yeah, I will. You know, like the script that I have is pretty large because I have it in like a GUI that I set all the parameters and started.

But I can make it like a shorter version so you can see all my function and the mass. And I will share it with you by the end of the day. You can just send me like to the cylinder.

Okay. Just with that. You give me the positions we are trying.

I will try to reproduce it. And you can send me like the tactile mass that you are getting. That I could get that like as a base and I can reproduce it.

No worries. Okay. Perfect.

I will send it to you. Thank you so much for letting me know that. No, thank you.

Because like those are the kind of things that like we are looking forward like to get like to know like something is happening. Or because sometimes my computer is working really nice. But when I pass it to another computer like the GPU is missing.

Sometimes it's the CPU working and it's like doing shit. But thank you so much for sharing that with me, Kudos. Yeah.

And you can share that with me. Of course. During the weekend for sure.

And as soon as I see why maybe it's not like going great, I could share with you like what we can do. Of course. Yeah.

Thank you. No, no. It's perfect.

And another thing is that I'm reading the tactile data, right? It's coming from the CNNs. Yes. But do we, I mean, is there any benefit, like is it good if I look into the formation files as well? Is it going to help me or we just, I can, there is nothing I can do with them? As long as they are going into the CNN and send me the 28 numbers, that's it? Yes.

It was more like in case you want to debug like something, but in general it doesn't, it's really not worth it. At one point just like making garbage and I think there is the option that you can cancel one, like you can cancel the formation because the files are really heavy. Okay.

Okay. No, I thought maybe I can look for the angle in that files, like if I find the angle of the tilt somehow by analyzing those files, but if, yeah, I think the only useful is the final data, tactile data that we get. Yeah, because honestly there you are going to be able to see it because it's more dense.

The tactile data we get from the sensor is like four columns, seven rows, so it's kind of limited. And the other ones is 18 per 12, so you have more visibility, avoidance are happening, so for sure you see more. But at one point I think your network or whatever you are using, it could confuse more if you are getting that because you are in a completely different like space resolution because it's in millimeters, really small formations.

So at one point I think it's going to be, for your algorithm, like at one point like more work to try to... I understand. To understand both. So, but yes, let me take a look to this and I'm going to keep you posted and I hope we can solve it so you can continue working with us.

Yeah, thank you so much. Thank you. I will share some stuff.

And all this stuff that we have and the new version that you're going to release, these are all for the old sensor. Is that right? Yes. Oh, which package do you don't know? Ah, because you have the old sensor, right? Yeah.

Oh, shit, yes. No, everything is coming from the old sensor and the new sensor. I don't know why I had the impression that it was with the new one.

No, it's with the old one. Yeah, yeah, yeah. So these are all for the old sensor? Yeah, everything is going to be for the old sensor and for the new one.

But in general, for the old one, because to be honest, it's a little more stable sometimes. Okay. But they're pretty similar, but at one point it's kind of complicated.

You know, in simulation, it doesn't matter for me as long as I get good data and I can use them. But once I move to the real robot, I need to know what is all of this. If the new sensor, I need to have the new sensor in the lab to be able to validate.

It's called the old sensor. Yes, that's really what you said to me because I had the impression that it was with the new one. But in general, I'm going to check everything that we're doing.

First, it's coming for the old sensor and then we are training the old sensor to adapt it to the new one. But basically, we need to do it for the old sensor because it's the one I'm using for my pushing. Right.

So, yes, it's going to be for that one too. Okay. Perfect.

Thank you so much. Thank you for your time. I will send some stuff for you.

Do you like it on the email or Teams? I can email you if you... Just on Teams, you can say the easiest way for you to know where it's live. Thank you so much. And whatever you see, whatever you find a problem, let me know.

That's pretty good to make a bet. Yeah, sure. Yeah, thank you so much.

We'll be in touch. Again, thank you for your time. Good luck.

Work hard. See you soon. See you.

Bye-bye. Take care. Bye.